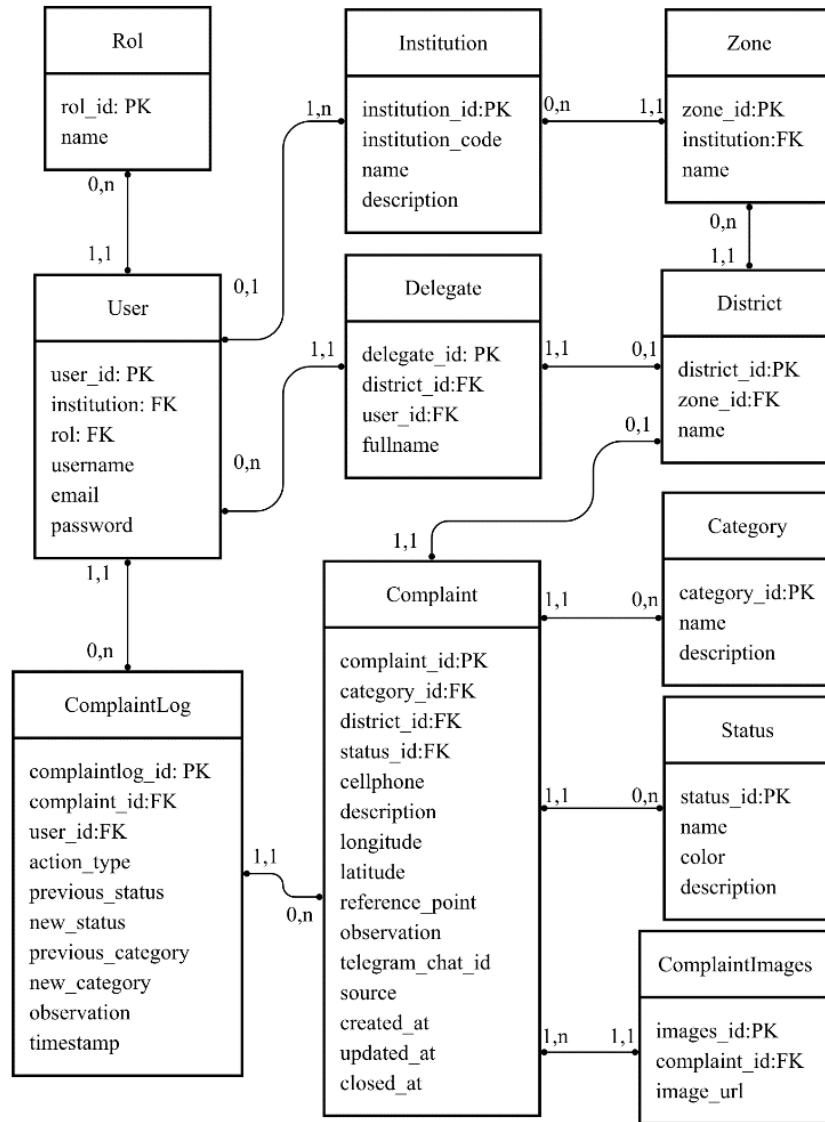


APPENDIX 2. System Requirements, Data Model, and Evaluation Protocol

A. ELICITED FUNCTIONAL REQUIREMENTS

ID	Description	Dependencies
R01	The system allows administrative users to authenticate with a username and password	-
R02	The system allows citizens to submit environmental complaints through a public Web form without authentication	-
R03	The system allows complaint submission via Telegram bot with a guided conversational flow	-
R04	The system validates mandatory complaint data before registration and requests correction if incomplete	R02, R03
R05	The system generates a unique alphanumeric tracking code for each registered complaint	R04
R06	The system assigns an initial status (<i>Pending</i>) upon successful complaint registration	R05
R07	The system supports hierarchical territorial selection (institution, zone, and district)	R02
R08	The system automatically routes complaints to the appropriate district delegate based on territorial selection	R07
R09	The system allows delegates to review, classify, and update complaint status	R01
R10	The system records all complaint modifications (status, category, observations) in the audit log	R09
R11	The system allows uploading multiple pieces of photographic evidence per complaint	R02
R12	The system provides a dashboard with aggregated statistics filtered by district, category, status, and date	R09
R13	The system notifies administrators and delegates upon new complaint registration	R06
R14	The system notifies citizens when the complaint status changes, according to the original submission channel (Web or Telegram)	R09
R15	The system allows citizens to consult the complaint status using the tracking code	R05

B. LOGICAL RELATIONAL DATABASE SCHEMA



C. TASK ASSESSED AND METRICS COLLECTED DURING CONTROLLED SIMULATION

Task	Role	Description	Metrics	Instruments
T1	Citizen reporter	Create a complaint via the web	Success/failure, completion time (s), errors	Observation + system logs
T2	Citizen reporter	Create a complaint via Telegram	Success/failure, completion time (s), errors	Observation + system logs
T3	Citizen reporter	Track complaint status	Success/failure, completion time (s), errors	Observation
T4	Delegate	Validate incoming complaint	Access time (s), errors	Observation + system logs
T5	Delegate	Update complaint category and status	Processing time (s), errors	Observation + system logs
T6	Delegate	Add observation and close the complaint	Success/failure, completion time (s)	Observation + system logs
T7	Admin	Review dashboard metrics	Completion time (s), errors	Observation
T8	Admin	Apply filters by district and date	Completion time (s), errors	Observation

D. ALIGNMENT BETWEEN RESEARCH QUESTIONS, EVALUATION FOCUS, TASKS, METRICS AND INSTRUMENTS

RQ	<i>Evaluation Focus</i>	<i>Tasks</i>	<i>Metrics</i>	<i>Instruments</i>
RQ1	System usability and citizen interaction	Execution of complaint-submission tasks by citizen reporters	SUS score, task completion rate, task success/failure, execution time, observed errors, perceived usability	SUS questionnaire, perception questionnaire, direct observation, task script
RQ2	Operational workflow efficiency and institutional feasibility	End-to-end execution of the complaint workflow across user roles	Completion times by task and role, number of completed steps, first response time, resolution time, workflow bottlenecks	Task timing records, observation sheets, workflow review, system logs
RQ3	Workflow consistency and multichannel integration	Multichannel complaint processing and cross-role workflow transitions	Consistency of state transitions, synchronization across channels, routing accuracy, correctness of role-based actions	System logs, observation, complaint trace review
RQ4	Technical and practical viability under realistic usage conditions	Full scenario execution and post-evaluation operation at current deployment scale	Infrastructure-related incidents, continuity of operation, observed technical failures, post-evaluation operational continuity	Deployment evidence, qualitative feedback, direct observation, technical validation records